

# A Comparative Study of Multi-modal Fusion Strategies for Fake News Detection

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**Abstract**—As there is a lot of misinformation on the internet caused mainly due to increasingly use of multi-model content generation using text and images. This paper will analyse the performance of different multi-model fusion strategies for fake news detection. We will use Fakeddit dataset to implement and compare 3 fusion methods: early fusion, late fusion, and cross-attention fusion. Text and image features will be embedded using pretrained models as Bert and Resnet50 respectively. Experimental results demonstrated that Early Fusion achieved the highest overall performance with an accuracy of 79.01% and F1-score of 79.03%, while Cross-Attention Fusion exhibited stronger sensitivity toward fake news detection. The study further identified significant over fitting across all models, stating the challenges of multi-modal representation learning under limited computational resources. Overall, the results suggested that simpler multi-modal fusion methods may generalise better than more complex attention-based architectures under constrained environments.

**Index Terms**—Early fusion, Late fusion, Cross-attention fusion

## I. INTRODUCTION

THE fake information spreads in text and images and detecting them could be challenging. The Text-only model ignores visual information and the Image-only model ignores important text. Fusion methods are still new and have an impact on how they perform on this task. [1] Most multi-modal fake news detection studies adopt a single fusion method, typically early fusion. This project systematically evaluates three fusion methods under an identical environment, isolating the effect of the fusion mechanism. It remains unclear whether simple fusion methods are sufficient or whether more sophisticated cross-modal interactions are necessary [2].

## II. CONTRIBUTIONS

- a) : Developed and evaluated three multi-modal fusion architectures for fake news detection.
- b) : Conducted a comparative analysis between Early Fusion, Late Fusion, and Cross-Attention Fusion strategies.
- c) : Demonstrated that simpler fusion methods achieved stronger generalisation performance than more complex attention-based architectures under constrained computational conditions.

## III. RELATED WORK

### A. Text-based models

Text-based methods are widely used due their amazing understanding of linguistic and semantics of text. Transformer based models, particularly BERT, perform extremely well and understand the relationship between words with its self-attention mechanisms. [3]

However, text-based model lack visual information which critical when dealing with multi-modal misinformation which may provide more context. Therefore, relying only on text based models is not enough in today's modern day of fake information detection.

### B. Image-based models

Convolutional neural networks (CNN's) are excellent in extracting visual features from an image. Architectures such as ResNet are used widely due to it learning hierarchical feature representations from visual data effectively. These types of model learn to catch patterns and visual inconsistencies that maybe manipulated. [4] [5]

However, as stated before, these models are limited to only images and lack explicit semantic meaning without any textual context. Thus, relying solely on image-based models can result in lower performance.

### C. Multi-modal Approaches

To overcome the constraints of uni-modal approaches, recent studies have investigated multi-modal algorithms that combine textual and visual information for the detection of fake news. These methods aim to learn the relation between images and text. The most common being: Early fusion, where usually the features extracted from images and text are combined and fed into a classifier. While this method is simple, it does not capture complex representation between each modality. Other advanced methods such as attention mechanisms or latent variable models try to model these cross-modal relationships. Multi-modal architectures using attention allow the model to focus on relevant parts of the image based on textual context.

Although we had these advances, many papers focused only on one fusion strategy [6] and failed to evaluate different fusion mechanisms under controlled conditions. Thus, it remains unclear whether simple fusion methods are sufficient or whether more sophisticated cross-modal interactions are necessary. This paper addresses this gap by systematically comparing Early Fusion, Late Fusion, and Cross-Attention Fusion using the same encoders, dataset, and training configuration. We also study whether complex attention-based architectures provide meaningful performance improvements over simpler multi-modal fusion techniques in resource-constrained environments.

## IV. METHODOLOGY

### A. Model Architecture

Our proposed model consists of 2 parts: a text and image encoder. These encoders give embedding for each modality and then are combined using different fusion methods.

a) *Text Encoder*: A pretrained transformer model, mBERT [7] (bert-base-multilingual-cased), is used to encode textual input. This is used due to its lightweight and efficiency while still maintaining strong performance. Given an input sequence, the model produces a contextual embedding:  $h_t \in R^{768}$

b) *Image Encoder*: A convolutional neural network based on ResNet [8] is used due to its ability to learn hierarchical features. Resnet extracts and learns visual features:  $h_i \in R^{2048}$

Our whole model structure and architecture is shown in figure1.

### B. Fusion Strategies

1) *Early fusion*: Early fusion [9] combines feature representations at the embedding level by concatenating the predicted features from the encoders:

$$h = [W_t h_t; ||; W_i h_i] \quad (1)$$

2) *Late fusion*: Late fusion [9] combines predictions from separate uni-modal models:

$$P = \alpha P_t + (1 - \alpha) P_i \quad (2)$$

3) *Cross-attention fusion*: Cross-attention [10] allows connection between modalities by allowing one modality to attend to another:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (3)$$

A visual architecture of fusion models are shown in sub figures 4a ,4b and 4c in Figure 2.

## V. DATASET & PREPROCESSING

We will use FAKEDDIT [13] dataset for our comparison. It is a large scale multi-modal dataset designed specifically for fake news detection. This dataset contains posts of Reddit with its text and related image, along with labels for each pair. Its multi modal nature makes it extremely suitable for evaluating fusion methods, as it provides aligned text-image pairs for each sample.

This is set as a binary classification problem, where posts are categorised as either real or fake. The dataset will be divided into training (80%), validation (6%), and test (14%) sets to ensure proper model training and evaluation. The dataset has 11167 images and text pairs with 5009 real and 6158 fake posts. This is a sample of FAKEDDIT dataset as shown in figure3.

a) *Preprocessing*: Text data is tokenised using a pretrained transformer tokeniser and image are normalised and resized to 224 x 224 pixels to match with pretrained ResNet weights [8]. Missing or corrupted image samples are removed and labels are converted into binary format.

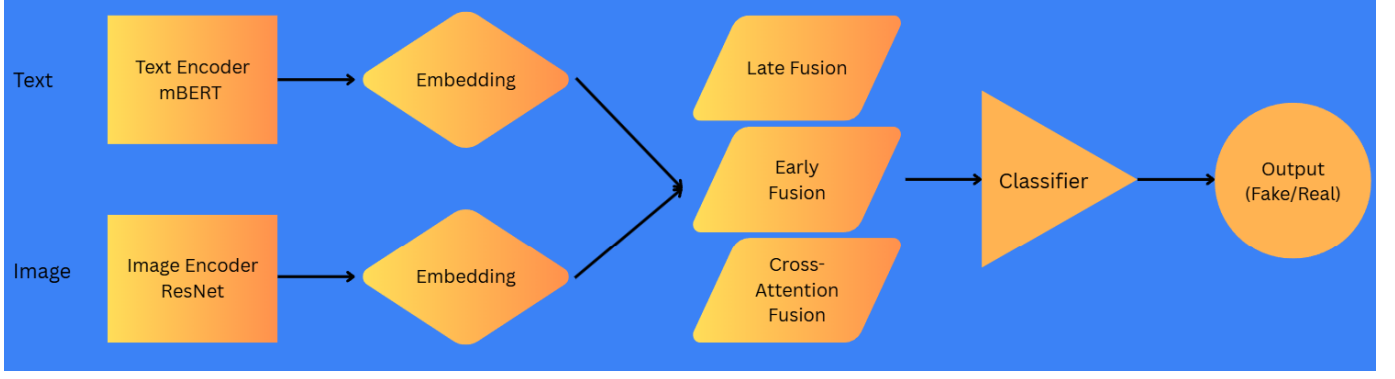


Fig. 1. Model Architecture.

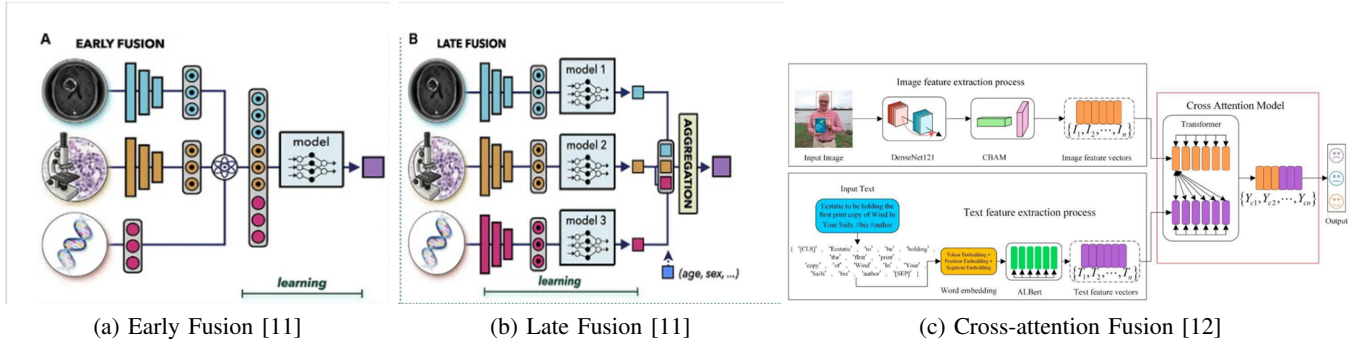


Fig. 2. All fusion model architectures.



Fig. 3. Sample from dataset.

TABLE I  
HYPER-PARAMETERS

| Parameter     | Value |
|---------------|-------|
| Epochs        | 15    |
| Batch Size    | 32    |
| Learning Rate | 1e-5  |
| Optimiser     | AdamW |

## VI. EXPERIMENT

### A. Hardware

Experiments were conducted using PyTorch (2.10.0+cu128) [14] on an NVIDIA RTX 5050 Laptop GPU with 16GB RAM.

### B. Training Pipeline & Hyper-parameters

The models are trained using binary cross-entropy loss:

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (4)$$

TABLE II  
PERFORMANCE COMPARISON OF MULTIMODAL FUSION STRATEGIES ON THE FAKEDDIT DATASET

| Method                 | Accuracy     | Precision    | Recall       | F1-Score     | AUC           |
|------------------------|--------------|--------------|--------------|--------------|---------------|
| Early Fusion           | <b>79.01</b> | <b>79.06</b> | <b>79.01</b> | <b>79.03</b> | <b>0.7888</b> |
| Late Fusion            | 78.89        | 78.85        | 78.89        | 78.85        | 0.7850        |
| Cross-Attention Fusion | 76.97        | 77.33        | 76.97        | 76.61        | 0.7587        |

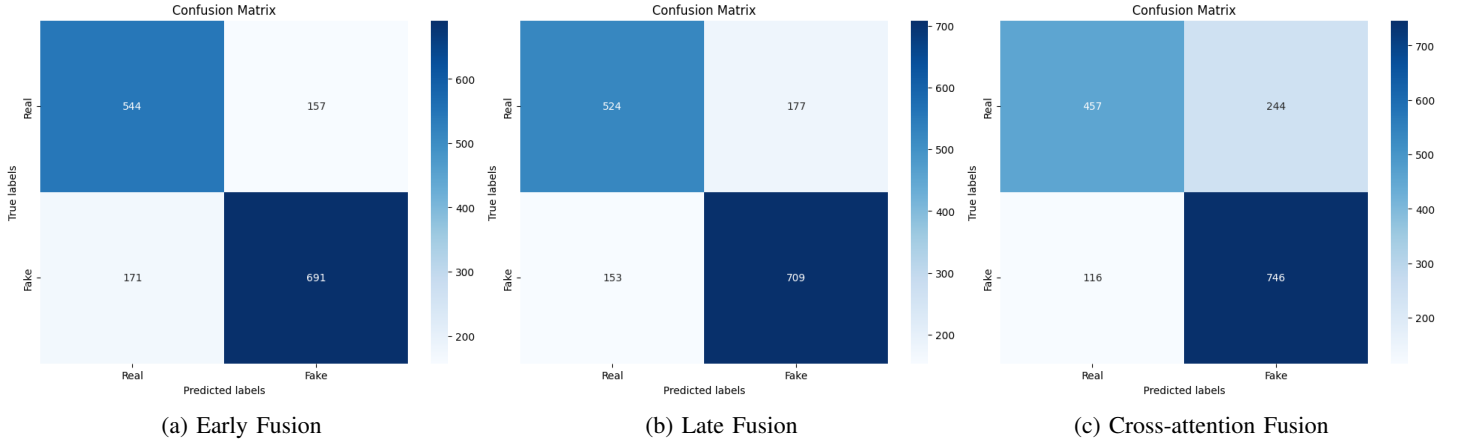


Fig. 4. All fusion model confusion matrices.

AdamW optimiser is used to optimise the training. Pretrained encoders are only used as encoders. Sigmoid Activation function is used. To ensure fairness across evaluation of each fusion strategies, we have same training epochs, dataset split and optimiser.

## VII. RESULTS

The performance of models were evaluated using multiple metrics to provide a extensive overview of classification performance. Metrics used were are: Accuracy, Precision, Recall, F1-Score and ROC-AUC.

Comparative results are shown in Table II.

As we can see Early Fusion had the best metrics demonstrating that it performed the best only when compared with these metrics. To learn more about the performance we have to look at their confusion matrices respectively.

The confusion matrices of each model are shown in Figure4.

## VIII. DISCUSSION

The results show that all three multi-modal fusion strategies achieved similar performance, as the test

accuracies ranged from 77% to 79%. From the evaluated methods, Early Fusion achieved the highest overall performance, obtaining a test accuracy of 79.01%, F1-score of 79.03% and an AUC score of 0.7888. This suggests that directly combining textual and visual embeddings was overall better for multi-modal fake news detection.

Late Fusion also achieved similar performance, with a test accuracy of 78.89% and F1-score of 78.85%. Although slightly lower than Early Fusion, its confusion matrix indicated more balanced predictions across both classes. This suggests that individually learning predictions before combining them, may improve model stability and reduce bias toward a particular class.

Cross-Attention Fusion produced the lowest overall accuracy and an AUC score among the all three approaches. However, the confusion matrix reveals that the model correctly identified a larger number of fake news samples, indicating a stronger sensitivity towards fake news detection. This suggests that attention-based interaction between textual and visual modalities improved the model's ability to learn subtle inconsistencies. However, due to its architectural complexity the risk of over fitting likely increased and reduced generalisation performance

under the our limited conditions.

### A. Limitations

Multiple limitations were found which affected the overall performance and were outside scope of this research.

Firstly, across all experiments there was a significant gap between training and testing performance. The training accuracies for all 3 models were close to 99%, while test accuracies were around 79% III. This indicated severe overfitting, showing that the models learned dataset-specific patterns instead of learning fully generalisable representations.

TABLE III

TRAINING AND VALIDATION ACCURACY OF FUSION MODELS

| Fusion Method   | Train (%) | Validation (%) |
|-----------------|-----------|----------------|
| Early Fusion    | 99.18     | 78.24          |
| Late Fusion     | 99.14     | 74.37          |
| Cross-Attention | 98.86     | 76.75          |

This issue was specifically noticeable in the Cross-Attention Fusion model as it has higher architectural complexity and increased parameter count. Attention-based models generally require larger datasets, stronger regularisation, and extensive hyper-parameter tuning to achieve stable generalisation performance. So it appeared to be most in Cross-Attention Model

Secondly, the experiments were conducted under limited computing resources using a single consumer-grade GPU. This limited the ability to train larger multi-modal architectures and models for a longer period of time.

Third, each experiment was trained for 15 epochs only due to computational limitations and training time considerations. Additional training and tuning could have improved model performance.

Finally, only 3 fusion strategies were evaluated. Other multi-modal learning approaches such as gated fusion, hierarchical fusion, or CLIP-based architectures [15] were outside the scope of this project.

### B. Future Work

These are some of my recommendations that could be used in future to improve multi-modal fake news detection performance.

Firstly, using stronger regularisation techniques [16] such as increased dropout, weight decay, and early stopping as it may reduce over fitting and improve generalisation performance.

Secondly, future work could also evaluate more advanced transformer-based multi-modal models such as CLIP [15], ViLT [17], BLIP [18], or multi-modal large language models. These models could improve cross-modal representation learning and learn better semantics between text and image modalities.

Third, using data augmentation [19] techniques may improve robustness. Image augmentation methods such as random flipping and rotation could increase visual diversity, while text augmentation methods such as synonym replacement or paraphrasing may improve textual diversity as well.

Finally, an increase in dataset size and computational resources will allow more complex attention-based architectures to achieve stronger performance.

### C. What I learned

Before attempting this experiment, I assumed that cross-attention fusion would have performed better due to its stronger theoretical modelling capabilities. However, I learned that increasing architectural complexity does not necessarily improve classification performance. I found that simpler fusion strategies may generalise more effectively under constrained training conditions.

Secondly, this project allowed me to really understand different types of metrics of evaluating machine learning systems rather than just accuracy. I learned the significance of using multiple metrics when evaluating as it provided deeper and better understanding of the model behaviour.

Overall, this project improved my limited understanding of multi-modal deep learning, fusion strategies, model generalisation, and the practical challenges involved in training large-scale multi-modal architectures.

## IX. CONCLUSION

In conclusion, an empirical comparison of fusion methods to show how each one performs was conducted. I conducted a comparative study of multi-modal fusion strategies for fake news detection using textual and visual information using the FAKEDDIT [13] dataset. Then 3 fusion models — Early Fusion, Late Fusion, and Cross-Attention Fusion — were implemented and then assessed under same experimental conditions using pretrained mBERT and ResNet encoders.

As results showed, Late Fusion had more balanced predictions and Cross-Attention Fusion had stronger sensitivity towards fake content. Early Fusion had the best overall performance based on the metrics. So these results show that simpler fusion methods generalise much better than complex attention-based methods under limited computing resources.

However, these models faced some limitations such as over fitting. Despite these limitations, using multi-model for fake news detection was effective and it provided valuable information and understanding between complex models, computational efficiency, and classification performance.

Overall, I hope that this research contributes to the field of multi-modal and fake news detection by providing a systematic comparative assessment of fusion strategies.

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